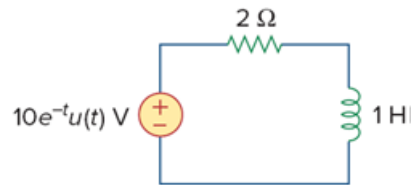


Problem

Find the energy dissipated by the resistor in the circuit of Fig. 18.53.

Figure 18.53



Step-by-step solution

Step 1 of 4

Consider the following circuit diagram:

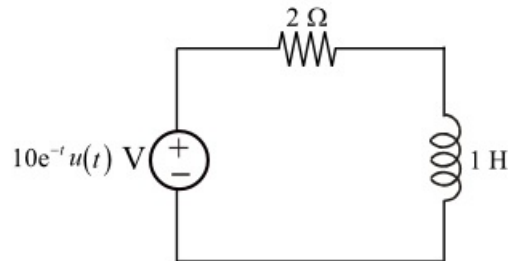


Figure 1

Step 2 of 4

Draw the Figure 1 in Frequency domain.

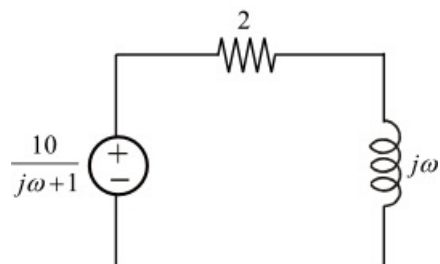


Figure 2

Step 3 of 4

Calculate the voltage across the 2Ω resistor using the voltage division formula.

$$\begin{aligned} V(\omega) &= \left(\frac{10}{j\omega + 1} \right) \left(\frac{2}{2 + j\omega} \right) \\ &= \frac{20}{(j\omega + 1)(2 + j\omega)} \\ &= \frac{A}{(j\omega + 1)} + \frac{B}{(j\omega + 2)} \end{aligned}$$

Here A and B are constants.

Calculate the constant value A , using the residue method.

$$\begin{aligned}
 A &= (j\omega + 1)V(\omega) \Big|_{j\omega = -1} \\
 &= (j\omega + 1) \left(\frac{20}{(j\omega + 1)(2 + j\omega)} \right) \Big|_{j\omega = -1} \\
 &= \left(\frac{20}{(2 - 1)} \right) \\
 &= 20
 \end{aligned}$$

Calculate the constant value B using the residue method.

$$\begin{aligned}
 B &= (j\omega + 2)V(\omega) \Big|_{j\omega = -2} \\
 &= (j\omega + 2) \left(\frac{20}{(j\omega + 1)(2 + j\omega)} \right) \Big|_{j\omega = -2} \\
 &= \left(\frac{20}{(-2 + 1)} \right) \\
 &= -20
 \end{aligned}$$

Substitute obtained A and B values in equation $V(\omega)$.

$$V(\omega) = \frac{16}{(j\omega + 1)} + \frac{-16}{(j\omega + 2)}$$

Apply the inverse Fourier transform to find $v(t)$.

$$\begin{aligned}
 v(t) &= \mathcal{F}^{-1}(V(\omega)) \\
 &= \mathcal{F}^{-1} \left[\frac{20}{(j\omega + 1)} + \frac{-20}{(j\omega + 2)} \right] \\
 &= \mathcal{F}^{-1} \left(\frac{20}{(j\omega + 1)} \right) - \mathcal{F}^{-1} \left(\frac{20}{(j\omega + 2)} \right) \quad \left(\because e^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a + j\omega} \right) \\
 &= 20e^{-t}u(t) - 20e^{-2t}u(t) \\
 &= 20(e^{-t} - e^{-2t})u(t) \text{ V}
 \end{aligned}$$

Thus, the voltage across the 2Ω resistor is $20(e^{-t} - e^{-2t})u(t)$ V.

Calculate the energy dissipated by the 2Ω resistor.

$$\begin{aligned}
 W &= \frac{1}{2} \int_{-\infty}^{\infty} v^2(t) dt \\
 &= \frac{1}{2} \int_0^{\infty} (20(e^{-t} - e^{-2t}))^2 dt \\
 &= \frac{1}{2} (400) \int_0^{\infty} (e^{-2t} + e^{-4t} - 2e^{-t}e^{-2t}) dt \\
 &= 200 \int_0^{\infty} (e^{-2t} + e^{-4t} - 2e^{-3t}) dt
 \end{aligned}$$

Simplify the equation further.

$$\begin{aligned}
 W &= 200 \left(\frac{e^{-2t}}{-2} \Big|_{t=0}^{\infty} + \frac{e^{-4t}}{-4} \Big|_{t=0}^{\infty} - 2 \frac{e^{-3t}}{-3} \Big|_{t=0}^{\infty} \right) \\
 &= 200 \left(\frac{1}{-2} (e^{-2(\infty)} - e^{-2(0)}) + \frac{1}{-4} (e^{-4(\infty)} - e^{-4(0)}) - 2 \left(\frac{1}{-3} \right) (e^{-3(\infty)} - e^{-3(0)}) \right) \\
 &= 200 \left(\frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right) \\
 &= 16.67 \text{ J}
 \end{aligned}$$

Thus, the energy dissipated by the resistor in the circuit in Figure 1 is 16.67 J.